USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22ECES501]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

ADVANCED SENSOR TECHNOLOGY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define Refractive index w.r.t optical fiber communication	L1	CO1	**02**
	b.	Define cardiac output	L1	CO2	**02**
	c.	Define bio medical electrodes	L1	CO3	**02**
	d.	What is electromagnetic interference (EMI).	L1	CO4	**02**
	e.	What are gyroscopes	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the working of Ion sensitive FET(ISFET) with neat sketch. what is the main challenge in designing ISFET.	L3	CO1	**05**
		OR			
	b.	Explain digital fiber -optic driver circuit	L3	CO1	**05**
3.	**a.**	Explain oscillometric method of measuring BP	L3	CO2	**05**
		OR			
	b.	Explain ultrasonic method of measuring BP	L3	CO2	**05**
4.	**a.**	What are polarized and non-polarized electrodes	L3	CO3	**05**
		OR			
	b.	Describe problems with surface electrodes	L3	CO3	**05**
5.	**a.**	What are the EMI problems in medical equipment	L3	CO4	**05**
		OR			
	b.	How the nuclear medicine equipment different from X -ray machine explain briefly	L3	CO4	**05**
6.	**a.**	Explain the principle of operation of Ring Laser gyro 5	L3	CO5	**05**
		OR			
	b.	Explain measurement of blood glucose by enzymatic approach using neat sketch.	L2	CO1	**05**
7.	**a.**	Briefly explain design requirement for an ideal blood gas probe.	L3	CO1	**05**
		OR			
	b.	List features of chemical fibro sensors.	L3	CO1	**05**

Deemed-to-be University Learning Resource Centre Jain Global Campus

		PART-C	**6 X 10=60**		
8.	**a.**	Explain pH meter with its construction and working.	L3	CO1	**10**
		OR			
	b.	Explain two stage transistor direct coupled amplifier laser diode driver circuit	L3	CO1	**10**
9.	**a.**	Explain instrumentation amplifiers with circuit and relevant mathematical equations	L3	CO2	**10**
		OR			
	b.	What is defibrillation? Explain any one defibrillator with circuit	L3	CO2	**10**
10.	**a.**	Explain spectrophotometers with their schematic diagram	L3	CO3	**10**
		OR			
	b.	Explain **i.** needle electrodes **ii.** indwelling electrodes **iii.** EEG electrodes	L3	CO3	**10**
11.	**a.**	Explain the operation of fluoroscopic machine with its block diagram	L3	CO4	**10**
		OR			
	b.	Explain three principal transmission paths for EMI into a sensor system	L3	CO4	**10**
12.	**a.**	**i.** Briefly explain intravascular measurement of oxygen saturation. **ii.** Explain reversible -dye type optical measurement of pH	L3	CO1	**5** **5**
		OR			
	b.	explain briefly different methods of measuring blood flow	L3	CO2	**10**
13.	**a.**	Explain preventive maintenance and corrective maintenance of medical equipment	L3	CO6	**10**
		OR			
	b.	Explain Path length control and Block material of RLG.	L3	CO5	**10**